

ChemCorp Green Chemistry Roadmap
2024-2030

CONFIDENTIAL — EXECUTIVE
COMMITTEE ONLY

EXECUTIVE SUMMARY

This document sets out the objectives and commitments of ChemCorp Industries for the period covered. It is intended for the Executive Committee and the Board of Directors. It serves as the reference for operational and strategic steering, KPI monitoring and stakeholder communication. Any modification must be approved by the CEO and the CFO. Document ref.: OBJ-VERT-2024-2030 — v2.0 dated 2024-02-28.

WHY A GREEN CHEMISTRY ROADMAP

European regulation (Green Deal, REACH, the green taxonomy), pressure from customers (buyer CSR criteria) and changing societal expectations place green chemistry at the heart of ChemCorp Industries' long-term competitiveness strategy.

Objective / Item	Target / Value	Comment / Detail
Pillar 1: Bio-based products	BioSolv range extended to 15 references by 2030	Replacing petrochemicals with renewable raw materials (agricultural biomass, waste)
Pillar 2: Clean processes	Energy -40% and water -30% by 2030	Heterogeneous catalysis (GreenCat), electrochemistry, continuous processes
Pillar 3: Circular economy	Used solvent regeneration service from 2026	"Solvent as a Service" model: collection, regeneration, redelivery
Pillar 4: Sustainable supply chain	100% of key suppliers CSR-assessed by 2026	EcoVadis score or equivalent required for suppliers above €200k per year
Pillar 5: Carbon transparency	Operational carbon neutrality (scope 1+2) by 2030	PAS 2060 certification targeted; residual offsetting via VCS

BIOSOLV PROGRAMME — DETAIL 2024-2026

The BioSolv programme is the flagship of the roadmap. It aims to substitute standard petrochemical solvents with functional equivalents derived from biomass.

Objective / Item	Target / Value	Comment / Detail
BioSolv-3 (methyl lactate acetate)	Commercial launch Q3 2024	97% bio-based, boiling point 144°C, KB solvency 48, ISCC+ certification Q4 2024
BioSolv-5 (technical limonene)	Launch Q1 2025	Derived from citrus peel (partnership with a southern cooperative), boiling point 176°C, "made in France" label
BioSolv-7 (bio-based glycol ether)	Feasibility study 2024, launch 2026	Co-development with IFPEN; potential 500 t/year in the electronics cleaning segment

Objective / Item	Target / Value	Comment / Detail
BioSolv-10 (degreaser range)	Launch 2026	Complete ready-to-use formulations — 4 references (light, standard, concentrated, high temperature)

GREENCAT PROGRAMME — GREEN CATALYSIS

GreenCat aims to develop reusable heterogeneous catalysts for ChemCorp's main synthesis reactions, reducing the consumption of homogeneous acids and the generation of salt.

Objective / Item	Target / Value	Comment / Detail
GreenCat-Est (esterification)	50 L pilot validated Q4 2024	Modified H-ZSM-5 zeolite; estimated energy saving -23%; ANR co-funding €180k
GreenCat-Hyd (hydration)	Laboratory study 2024-2025	Applied to the hydration of light olefins; partnership with CNRS Lyon
GreenCat-Ox (mild oxidation)	Exploratory research 2025-2027	Replacing KMnO4/CrO3 with Au/TiO2 catalysis for primary alcohols

KEY MILESTONES AND INDICATORS

The roadmap is steered through quarterly milestones and key performance indicators (KPIs) approved by the Executive Committee and the Board of Directors.

Objective / Item	Target / Value	Comment / Detail
Bio-based share of revenue	2024: 2% 2026: 12% 2030: 25%	Monthly monitoring — Executive Committee reporting
CO2 intensity (kgCO2 per tonne of product)	2024: -10% 2027: -30% 2030: -50%	2021 baseline; scope 1+2; annual Deloitte audit
Energy intensity (kWh per tonne)	2024: -8% 2027: -22% 2030: -40%	Monitored by workshop; real-time energy dashboards
Waste to landfill	2024: below 5% 2030: 0%	Zero non-recovered waste plan; partnership with Veolia since 2024
Green Chemistry R&D budget	2024: €380k 2025: €650k 2027: €1.2M	Of which 30% grant-funded (ANR, ADEME, Horizon Europe)

FUNDING AND GOVERNANCE

The roadmap draws on mixed funding (equity, green debt, grants, research tax credit) and a dedicated governance structure.

Objective / Item	Target / Value	Comment / Detail
Total budget 2024-2030	€14.2M (7 years)	40% equity, 25% BPI green debt, 20% grants, 15% research tax credit
Green Chemistry Steering Committee	Quarterly meeting	CEO + CTO + COO + CFO + Head of CSR + 1 independent director
External partners	Bureau Veritas (ISCC+) + Deloitte (ESG audit)	Reports published annually on the ChemCorp website
UN Sustainable Development Goals addressed	SDG 9, 12, 13, 17	Industrial innovation, responsible consumption, climate action

PRINCIPAL RISK ANALYSIS

Identified risk	Likelihood	Impact	Mitigation plan
Biomass availability / price	High	High	Multi-year supply contracts with cooperatives
Delay to ISCC+ certification	Medium	Medium	Weekly follow-up with Bureau Veritas; fallback: RSPO certification
Failure to secure ANR GreenCat funding	Low	High	Alternative funding via BPI France ETI Ready planned
Fast-moving competition in bio-based products	Medium	High	Accelerated patent filing + exclusivity for pilot customers
Bio-based content not recognised by regulation	Low	Medium	Participation in AFNOR and ChemBioEurope working groups

GOVERNANCE AND MONITORING ARRANGEMENTS

- » Green Chemistry Steering Committee (quarterly) — CEO + CTO + CFO + CSR
- » Monthly progress report to the Executive Committee
- » Public annual report aligned with GRI + SASB (published in March)
- » Annual external ESG audit (Deloitte) since 2022
- » Stakeholder dialogue (customers, suppliers, NGOs) — annual ChemCorp Vert forum

APPROVAL AND SIGNATURES

Prepared by	Reviewed by	Approved by
Dr. Martine Chabrol R&D Director	Ms Hélène Marchand Operations Director	Mr François Lemercier CEO
Signature:	Signature:	Signature:
Date:	Date:	Date: